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Geologic and Paleomagnetic Study of Permo-Carboniferous Redbeds (Sabaneta and Mérida Facies), Venezuelan Andes[1]

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Abstract Late Paleozoic redbeds of the Sabaneta and Mérida facies (Venezuela) are separated unconformably from redbeds of the Mesozoic La Quinta Formation. The older redbeds and volcanics associated with the Sabaneta facies have been sampled for paleomagnetic study with the following results.

1. No statistically significant data were obtained from the eight basalt samples despite alternating field (AF) demagnetization up to 700 oe.

2. Magnetic vectors of the seven Mérida redbed samples after thermal demagnetization to 550°C (D = 358, I = + 10, k = 6) resemble the present magnetic field, suggesting Quaternary-Holocene acquisition of remanence.

3. Twenty-two redbed samples from the Sabaneta facies, after thermal demagnetization to 550°C, give moderately consistent paleomagnetic vectors (D = 223, I = + 15, k = 11), which are best interpreted as of secondary origin acquired after Late Permian folding. If this secondary remanence is considered to be Late Permian, it implies a post-Permian 60° clockwise rotation of the Venezuelan Andes for which there is no corroborating paleomagnetic or tectonic evidence. Disregarding polarity, the Sabaneta magnetic axis closely resembles that for the La Quinta Formation in the type area which was acquired during Tertiary Andean uplift. It is concluded that the Sabaneta remanence was acquired contemporaneously in Tertiary time. Similarly, the northeast-southwest declination is anomalous and a clockwise tectonic rotation within the Venezuelan Andes is suggested in possible explanation.

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Introduction

One of the major unconformities in the Venezuelan Andes separates Mesozoic formations from the underlying late Paleozoic section. Recently we (Hargraves and Shagam, 1969) reported on the paleomagnetic characters of the La Quinta Formation, the basal unit of the section above the unconformity. The Sabaneta and Mérida facies, subjects of this paleomagnetic study, lie below the unconformity in association with Permo-Carboniferous marine shale and limestone.

The natural remanent magnetism (NRM) of the La Quinta rocks was concluded to be of secondary origin, acquired during The Tertiary uplift of the Andes. Measured declinations appeared to be about 20° east of that of the present dipole field. The only suspected primary magnetization measured was obtained from samples of tuff near the base of the section. The purpose of this study is to compare the paleomagnetic characteristics of the Sabaneta and Mérida rocks with those of the La Quinta and to check the tentative tectonic conclusions drawn from the results of the La Quinta study.

The specimens examined in this study were collected by Shagam in the summers of 1967 and 1968 in the course of regional field work in the Andes under the aegis of the Dirección de Geología, Ministerio de Minas e Hidrocarburos, Venezuela. The magnetic measurements were made at the Rock Magnetism Laboratory at Princeton University under the supervision of Hargraves and, in the latter's absence on sabbatical leave, F. J. Vine.

Abbreviations used in this paper are the following.

NRM	Natural remanent magnetization
J	Intensity of magnetization
n	Number of specimens
D	Declination of direction of magnetization
I	Inclination of direction of magnetization
k	Fisher dispersion parameter (Fisher, 1953)
α95	Cone of confidence about direction of magnetization at 95 percent level of probability
AF	Alternating field
oe	Oersted

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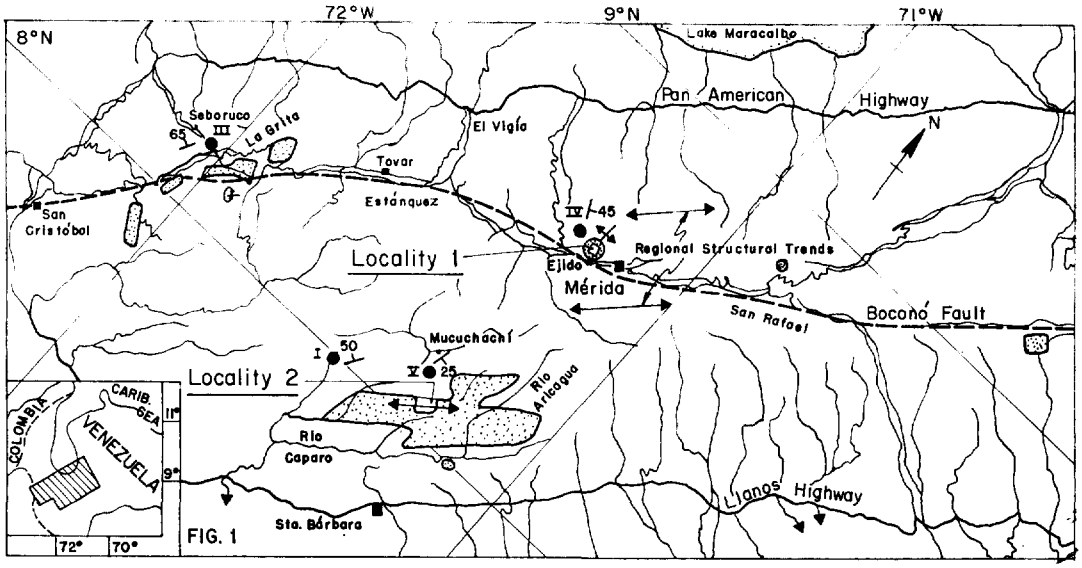


FIG. 1.—Generalized distribution of Permo-Carboniferous redbeds, Venezuelan Andes. Note bimodal distribution with concentration of localities along Boconó fault and extensions (dashed line). Sample localities 1 and 2 shown clear, others stippled. Mesozoic La Quinta redbed localities sampled by Hargraves and Shagam (1969) shown black. Principal structure of sampled localities shown; note contrast with regional structure at Mérida site.

Note: k is the best calculated estimate of K, the theoretical dispersion parameter. All values of K in Hargraves and Shagam (1969) should have been given as k.

General Geology

The distribution of the redbeds believed to be of late Paleozoic age is shown in Figure 1. The isolated outcrop areas may be grouped roughly into two separate (geographic and geologic) associations. The work of early investigators is summarized by Rivero (1956). The most recent and by far the most detailed study is that of Arnold (1966). Contemporaneously, additional studies have been made by W. R. Smith (unpub. reports, Creole Petroleum Corporation) and by Shagam (in prep.).

Recognition

Much the same problem of recognition of the redbed formations exists as for the La Quinta Formation (Hargraves and Shagam, 1969, p. 538). By tacit agreement redbeds known or believed to lie conformably below the fossiliferous marine Palmarito Formation of probable Early Permian age are assigned to the Sabaneta Formation. Such decisions may be complicated by the absence in outcrop of the Palmarito section (*cf.* Arnold, 1966, p. 2371, who considered one redbed section he examined to consist of a basal Sabaneta portion overlain by an upper La Quinta portion). In the vicinity of the Rio Caparo (Fig. 1) redbeds are associated with early Paleozoic rocks in possible imbricate thrust slices and conceivably could constitute an older, pre-Sabaneta, redbed formation (Shagam, 1968, p. 178).

On the basis of extensive regional mapping in the Central Andes, Shagam concluded that late Paleozoic relations are as depicted diagrammatically in Figure 4a. The Palmarito facies (south) is characterized by a rich and varied faunule and absence of metamorphism. The Palmarito facies (north) is distinctly metamorphosed and fossils, although clearly of Permo-Carboniferous aspect, are uncommon and deformed. Clearly, stratigraphic correlation of the two facies is by no means certain. In turn, correlation of northern redbeds (Mérida, Casés) with the southern (Sabaneta, Aricagua) is correspondingly uncertain.

The situation is further complicated by doubts concerning "established" Palmarito sections. Kovisars (1969) has shown that one such section exposed at Pico el Aguila, 50 km northeast of Mérida, is significantly different in lithology and stratigraphy from the type section. No fossils were found, and the section is intensely deformed and metamorphosed (staurolite was identified locally). Another section exposed on Cerro Azul, Barinas State, and con-

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Table 1. Comparison of La Quinta Formation and Sabaneta/Mérida Facies

Criteria		La Quinta	Sabaneta	Mérida
Color of redbeds		Red-purple	Grayish red	Grayish purple
Stratification		Well defined, laminated to massive; uniform, laterally extensive	Very poor; markedly lenticular	Very poor; markedly lenticular
Petrographic characters of sandstones and siltstones	Quartz	Rounded grains with abundant secondary silica overgrowths	Angular grains; rare secondary silica overgrowths	Angular grains; no secondary silica overgrowths seen
	Opaque minerals (all hematite)	Subrounded, concentrated parallel bedding, cross-bedding; clastic history	Irregular, hackly shape; not concentrated parallel bedding; diagenetic origin(?)	Angular; diagenetic (?)
	Carbonate	Matrix locally calcareous; no nodular limestone known	Rare aph. limestone lenses; abundant aph. limestone nodules in upper member. Matrix locally calcareous	Nodular aph. limestone not common; matrix locally calcareous
	Matrix	Commonly 10–15 percent; significant clay content	Commonly 5–10 percent; minor clay content	Commonly 15 percent; abundant clay content
	Texture	Poor intrabed sorting; subrounded clastic grains	Very poorly sorted throughout; microbreccia	Very poorly sorted throughout; microbreccia
	Cement	Generally well cemented; silica	Poorly cemented; friable; disintegrates in storage	Poor-moderate; less friable than Sabaneta
Pebbles-boulders		Rounded; concentrated in conglomerate layers	Rounded; rare conglomerate; mainly scattered in conglomeratic sandstone	Dominantly angular; in thick breccias
Metamorphism/deformation		Unmetamorphosed; tilt blocks; gentle folds	Minor stretching of quartz grains; intensely folded	Slaty-phyllitic in some outcrops; intensely folded

sidered (*cf.* Alberding, 1956, p. 133) to be possibly correlative with the Palmarito section, has been shown by Schubert (1969) to be of probable early Paleozoic age.

In view of these uncertainties we refer to late Paleozoic sections in terms of facies rather than formations. Sabaneta facies refers to the redbeds in the south-central Andes, considered to be overlain conformably by the Palmarito facies (south); Mérida facies (originally Mérida Formation; Kündig, 1938) refers to the redbeds in the Rio Chama valley, considered to be overlain conformably by the Palmarito facies (north). Additional reasons in support of the nomenclature employed are presented in subsequent discussion. Criteria listed in Table 1 are believed to be significant, when applied statistically, in distinguishing between Sabaneta, Mérida, and La Quinta redbeds in those areas where stratigraphic markers (Palmarito facies, Cretaceous formations) are not present.

Lithology and Stratigraphy

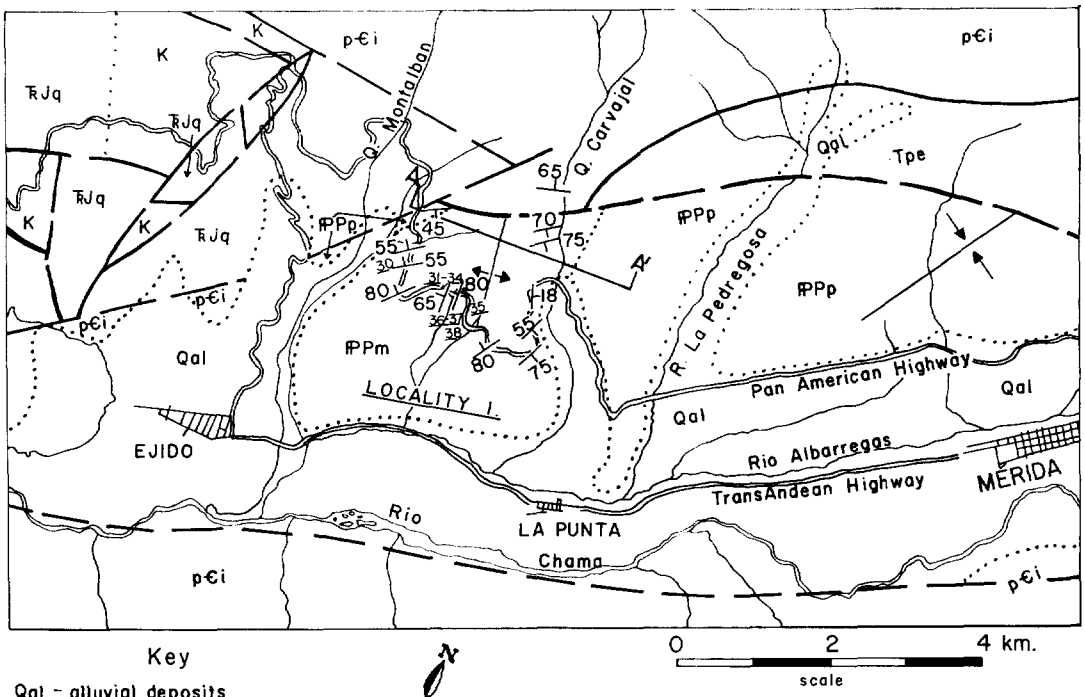
For the general characteristics of lithology and stratigraphy the reader is referred to the work of Arnold (1966) who measured sections in both the Sabaneta and Mérida areas. Shagam's data correspond closely with Arnold's for the Sabaneta section but differ significantly for the Mérida section. From Arnold's description (1966, p. 2371) it appears that he did not recognize the tightly folded anticline mapped by us (see section on structural geology, and cross section in Fig. 2). Hence it appears that his measured section is twice the correct thickness and the bottom half is inverted.

The specimens selected for paleomagnetic study were collected from the upper member of the Sabaneta facies and mainly from the lower part of the Mérida facies. The locations of specimens and other field data appear in Figures 2 and 3.

Most of the specimens from the Mérida facies (locality 1) are variegated, pale grayish purple to pale grayish green. Bedding is poorly developed but readily definable in most hand specimens and thin sections. Graded bedding was observed in two specimens. The specimens range from clayey siltstone to sandy and granule breccia, all characterized by wacke-type textures. The dominant clastic constituents are angular quartz grains, high-grade metamorphic rock fragments and slate chips. Minute flakes of white mica and chlorite are abundant. Angular grains of hematite occur in accessory amount.

The Sabaneta specimens (locality 2B) are dominantly grayish red and most are silty sandstone. Primary stratification is detected only with difficulty in large hand specimens and is virtually indefinable in most thin sections. Angular, embayed, clastic grains ($\geq$ 80 percent by volume) are composed dominantly of quartz with lesser amounts of turbid reddish-brown plagioclase and potash feldspars (commonly 20–30 percent of the clastic grains), and flakes of muscovite and chlorite. Textures range from wacke-type with about 15 percent matrix to micro-breccias displaying small isolated triangular areas of matrix and cement. In most specimens

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Key

Qal – alluvial deposits
Tpe – Tertiary Formations
K – Cretaceous Formations
TrJq – La Quinta Formation
PPp – Palmarito Facies
PPm – Mérida Facies
p€i – Iglesias Complex
30-38 – location of oriented samples
– strike & dip of beds
– fold axes

CROSS-SECTION A–A'

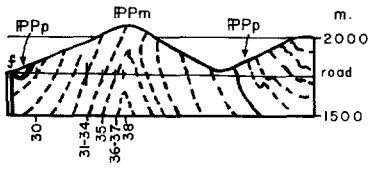


FIG. 2

Tentative compound stratigraphic column based on lithologies from W and E limbs of postulated anticline.

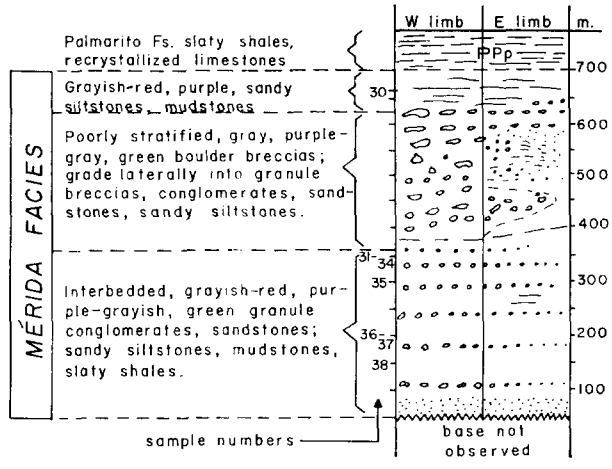


FIG. 2.—Mérida facies, sample locality 1. Stratigraphic column and anticlinal structure tentative. La Quinta locality IV (Hargraves and Shagam, 1969) is along road at top left corner of map.

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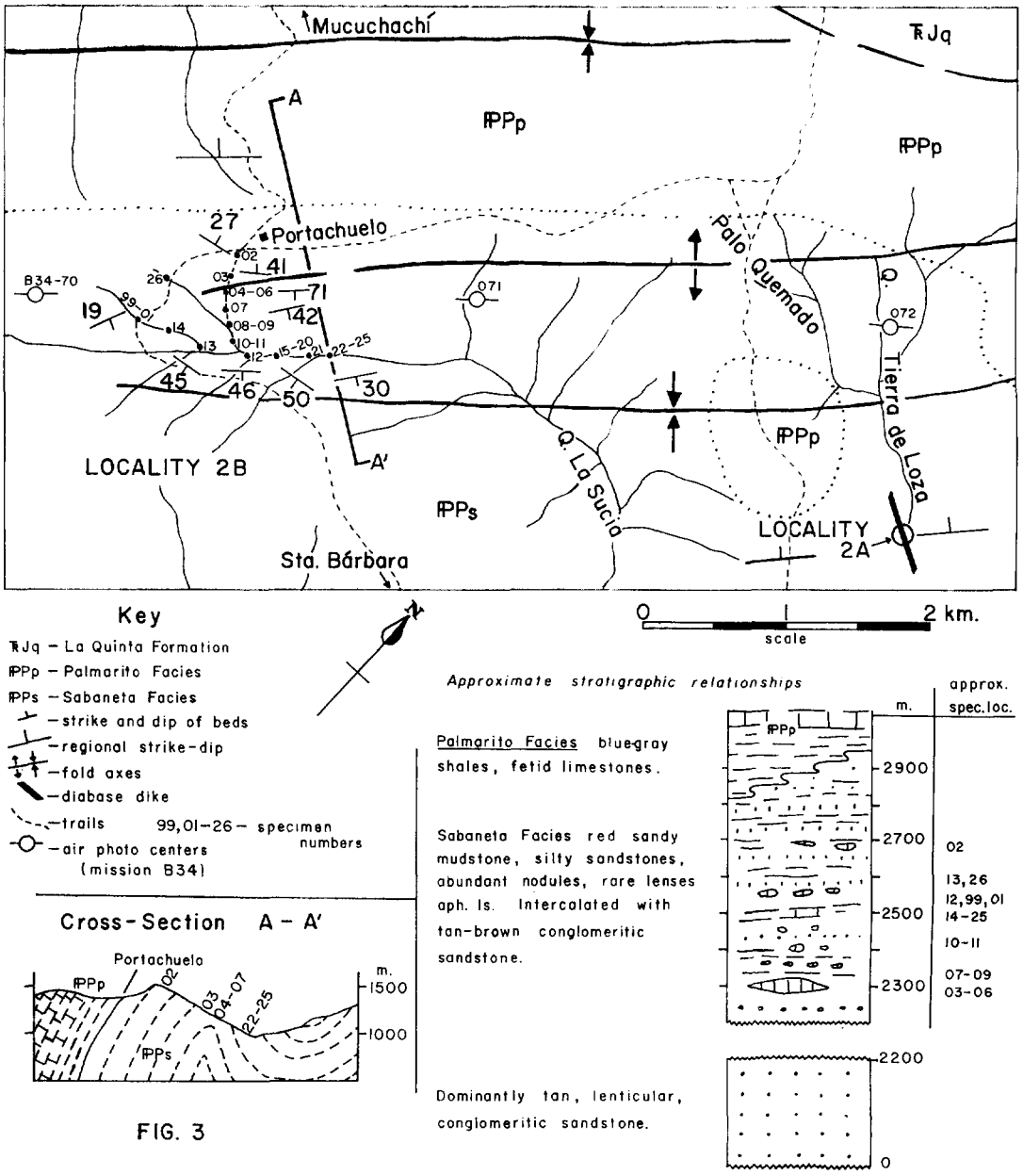


FIG. 3

FIG. 3.—Sabaneta facies type area, Mérida State: sample localities 2A, B. Absolute and relative stratigraphic locations of samples very approximate.

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the texture is modified by the presence of intraformational red mud chips, oval patches of finely crystalline calcite, or a blotchiness defined by color and possibly related to the activities of burrowing organisms. In many specimens micas and chlorite are distinctly oriented in approximately orthogonal conjugate planes which are approximately bisected by the normal to bedding. It is not known whether this is a compaction feature or related to secondary structural deformation.

Opaque grains are accessories in all thin sections. H. J. Dick took X-ray powder photographs of the opaque grains from three specimens and they proved to be α-hematite. In thin section these grains have a markedly hackly and serrate appearance and display irregular shapes (branching in some examples) suggesting that they are either not clastic, or that they represent clastic grains considerably modified and resorbed by postdepositional processes. In marked contrast to the La Quinta redbeds (Hargraves and Shagam, 1969, p. 540) a reddish hematitic scale on clastic grains is not a conspicuous feature and secondary silica overgrowths on clastic quartz grains were rarely observed. The relatively drab, grayish-red color of most specimens appears to be related in part to red-stained matrix and in part to reddish-brown weathered clastic feldspar grains, being lighter in tone in those specimens with little matrix.

The vesicular basalt specimens (locality 2A) were collected from a dike 2 m wide intruded into the Sabaneta facies. The volcanic rocks also are present as a few scattered, thin lava flows in the formation. In thin section these specimens display a sub-ophitic texture, but the most notable feature from a paleomagnetic point of view is the fact that augite is largely altered to amphibole or chlorite and plagioclase is markedly turbid. These effects are believed related to deuteric alteration.

Depositional Environment

We agree in general with the views of Arnold (1966, p. 2373) concerning the depositional environments of the Mérida and Sabaneta facies. The former appears to represent an alluvial fan formed at a fault scarp and having a northerly provenance; the latter represents a deltaic deposit derived from a southerly provenance.

Age

On the basis of pollen types, Arnold (1966, p. 2372) concluded tentatively that most of the Sabaneta Formation (Arnold's terminology) is of Permian age and ". . . that the lower part may be older (Carboniferous). . . ." On the basis of the faunas in the Palmarito Formation (Arnold's terminology) he concluded (1966, p. 2378) that it is probably of Early Permian age and that it might extend into the late Carboniferous. In the south-central Andes the contact between the two formations is gradational. On the basis of their spatial relations it appears more probable that the gradation is largely vertical rather than lateral. Pending confirmation, we consider the assigned faunal age for the Palmarito rocks more reliable than the pollen age for the Sabaneta facies, and therefore the latter is more likely of Carboniferous (Pennsylvanian?) age, possibly extending into the Early Permian.

The Mérida facies appears to lie conformably below a shale-limestone sequence of Palmarito aspect tentatively dated by Pierce *et al.* (1961), on the basis of a sparse fauna, to be of Permo-Carboniferous age. As implied by Arnold (1966), the two formations may be correlatives of the corresponding two facies in the south-central Andes. However, on the strength of the metamorphic evidence, it is conceivable that the Mérida-"Palmarito" association may be older than the Sabaneta-Palmarito pair. At present, the best conclusion is that the Mérida facies is of Permo-Carboniferous age possibly coeval in part or whole with the Sabaneta facies.

Tectonic Setting

The contrasting views of Arnold and Shagam on the tectonic setting of the Sabaneta/Mérida facies are depicted diagrammatically in Figure 4. The difference centers largely on the postulated tectonic environment of the Mucuchachí facies. Arnold (*in* Compañía Shell and Creole, 1964; 1966, p. 2382–2383) viewed the latter as a trough facies laterally equivalent to a Siluro-Ordovician shelf facies on the south. We favor an alternative view proposed by Smith (in Compañía Shell and Creole, 1964), that these are two unrelated sedimentary cycles, and further postulate that the Mucuchachí is a lateral facies equivalent of the Sabaneta facies. In support of this postulate is the fact that all faunas, although sparse, collected from the Mucuchachí slates appear to be of Permo-Carboniferous age.

Within the overall tectonic setting proposed here there appears to be a distinctive difference between the tectonic control of deposition of the Mérida facies (and allied deposits of the

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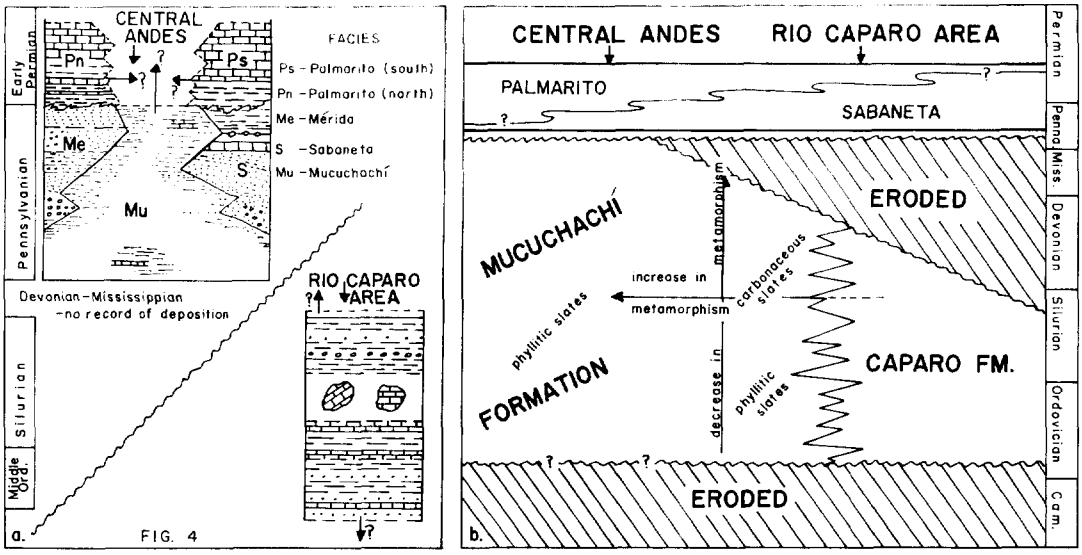


FIG. 4.—Contrasting views on tectonic setting of Sabaneta-Mérida facies: (a) the writers, (b) Arnold (in Compañía Shell and Creole, 1964; 1966). In view a axis of marine Mucuchachí basin was about halfway between Mérida and outcrop area of Sabaneta facies. Northern margin of basin was modified by narrow structural trough in which Mérida facies was deposited. For simplicity, mixed facies (Casés in north, Áricagua in south) intermediate between redbed and marine facies are not shown.

northern belt) and the Sabaneta facies. The former deposits are associated with a major fault belt which defines a graben or other structural trough roughly coincident with the axis of the mountain belt (Fig. 1). This structural trough appears to have been reactivated at least three times subsequently. In the south-central Andes there may have been some structural control of deposition of the Sabaneta facies, but it was far less pronounced and the sediments were not confined within a narrow trough. The difference in the two environments is markedly reflected in the lithologic and stratigraphic characters of the two formations as described by Arnold (1966). Therefore, it is unlikely that the two formations were ever joined as a continuous sheet, but the question serves to underscore the problems of recognition and age.

Structure

The Sabaneta and Mérida facies were folded tightly (together with the Palmarito and correlative formations) before La Quinta deposition (probably Middle to Late Permian). Post-Paleozoic deformations took the form of uplift and tilting dominantly, involving at least two "pulses" (*cf.* Hargraves and Shagam, 1969, p. 544).

In the Late Permian deformation the Sabaneta and Palmarito facies were deformed predominantly by flexural slip. Hence flowage is not believed to have had a significant effect on the orientation of magnetic vectors. Locally, a spaced (commonly ± 1 cm apart) "fracture" cleavage is developed in the micaceous sandy siltstone, and in cross section displays a pronounced sigmoidal shape related to the flexural slip. Fold axes trend northeast-southwest and are approximately horizontal (Fig. 3).

Because stratification is even harder to detect in outcrop in the Mérida facies than in the Sabaneta outcrop area, secondary structures such as folds are correspondingly more difficult to recognize. Field mapping suggests that the Mérida facies constitutes a tightly folded anticline with steeply dipping limbs (±60°). This raises the question of the correct determination of "tops" for the paleomagnetic study. G. D. Klein examined all specimens and made top determinations on the basis of sedimentologic criteria. In all cases they conformed to the top orientation selected in the field and were in agreement with the postulated anticlinal structure. In three specimens unequivocal top criteria were detected. The distribution of the Palmarito rocks relative to the beds also conforms with the anticlinal structure. On this basis there is, as previously mentioned, the possibility that Arnold's section is partly inverted and roughly

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twice the correct thickness (*cf.* Fig. 2). We made no attempt to correlate paleomagnetic characters with stratigraphy, hence the preceding comments are considered significant only insofar as they bear on top determinations. As indicated, sedimentologic criteria support the field observations.

The post-Paleozoic tilting also is significant only insofar as it may have affected the orientation of paleomagnetic vectors. As shown in Figure 5, on the basis of the space problem and the nature and intensity of secondary structural features in the Permo-Carboniferous section, it is reasonable to postulate that the post-Paleozoic formations were draped over a vertically rising core of late Paleozoic rocks; *i.e.*, that the latter were unaffected (or only slightly affected) by the post-Paleozoic tilt movements. Hence no significant rotation of magnetic vectors is required to obtain late Paleozoic orientations.

In the Mérida area the late Paleozoic section is in a structural trough. Regional mapping has made it clear that the fault-bound structural blocks north of the trough have been tilted northward. The Sierra Nevada on the south appears to have been tilted southward though the evidence is not unequivocal. The structural trough itself may possibly constitute a "keystone" graben of the type not uncommon along the axis separating blocks of opposite tilt (*cf.* Wise, 1963, p. 588–589). In this tectonic setting too, post-Paleozoic tilting need not have affected significantly the orientation of structural features formed in the late Paleozoic deformation, inasmuch as movement of the graben would be dominantly vertical.

There is an additional structural feature in the Mérida area which requires consideration. Fold axes in the late Paleozoic section diverge notably from those in Paleozoic and Precambrian (?) rocks in the regions on the north and south (Fig. 1). The trends of the former suggest the presence of a stress field dominated by a right-hand shear couple operating along the boundary faults of the graben. However, regional field studies do not entirely support such a possibility. The southern boundary fault is

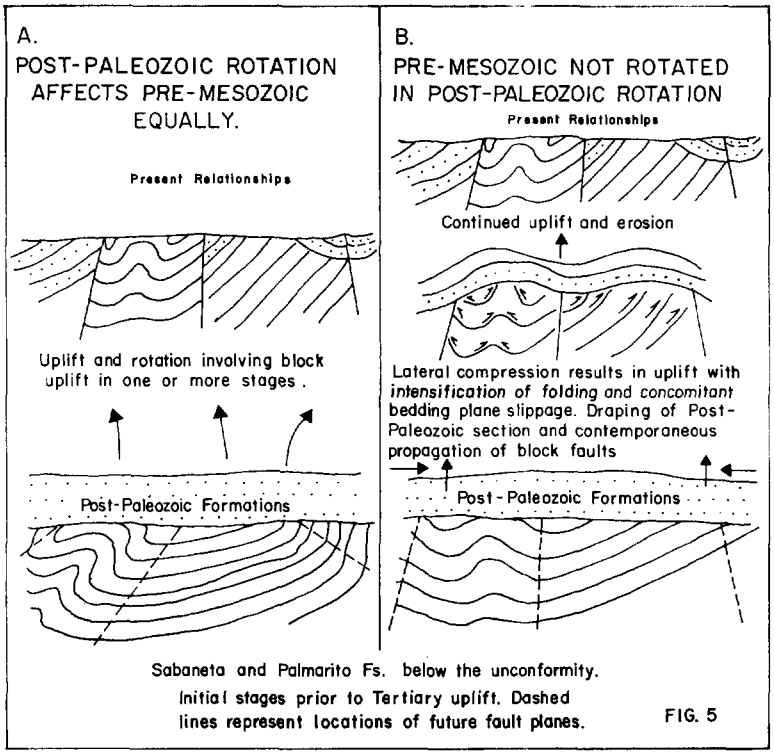


FIG. 5.—Diagrammatic representation of mechanics of Tertiary uplift showing Sabaneta facies (*A*) tilted through same angular distance as post-Paleozoic formations, and (*B*) preferred postulate: relatively unaffected by tilting.

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the continuation of the Boconó fault, believed by many (*e.g.*, Rod, 1956) to be a right-lateral strike-slip fault. We believe this fault is dominantly of high-angle reverse type with a minimal component of horizontal movement. The northern boundary fault appears to be normal, and there is no field evidence of a strike-slip component of movement. Directly west of the area of outcrop of the Mérida facies the post-Paleozoic section is in a series of small fault blocks that appear to represent gravity-slide blocks originating north of the graben. The structural characteristics are quite distinct from those of the late Paleozoic section, however, so an explanation involving gravity sliding for the divergence of the fold axes is not convincing. This problem is of significance in that the present orientation of these folds may have involved angular rotation about a vertical axis, from an original more northeasterly direction. This matter is further discussed in the light of the paleomagnetic results.

Field and Laboratory Methods

The oriented samples from the Mérida facies (locality 1) were collected from deep roadcuts that were excavated with explosives at least 7 years prior to sampling. The area is one of steep slopes and high relief marked by small landslides and slumps. As far as could be determined in the field, all the samples come from stable *in situ* outcrops. Absolute stratigraphic location shown in Figure 2 is very approximate, but relative stratigraphic position is considered reliable.

Locality 2B (Sabaneta facies) covers the northern half of the Quebrada La Sucia valley. In view of the widespread outcrops sampled, the high relief, and the lack of adequate topographic data, the absolute and relative stratigraphic locations of samples shown in Figure 3 are unreliable. Most specimens were collected from stream outcrops which appeared to be stable and *in situ*, but the principal problem concerns accurate definition of bedding (*cf.* description of lithologic characters above). Specimens were collected only where adequate evidence of bedding orientation in the outcrop was observed. A few specimens were collected on interfluves or trails and these are the most likely to have suffered rotation as a consequence of surface slumping.

Twenty-nine of the 35 redbed specimens collected, and the 8 basalt samples, survived transport and storage. Cores 0.98 in. in diameter were drilled from sample in a drill press; the samples and cores were reoriented in a sandbox. Three specimens, 1 in. long, were cut from each redbed core (only two from the basalt), and used for the magnetic measurements.

The natural remanence was measured on a Princeton Applied Research model S.M. 1, spinner magnetometer (2-axis tumbler with inductrol). Thermal and alternating field demagnetization was performed on standard equipment as described by Hargraves and Shagam (1969, p. 545).

Paleomagnetic Results

Volcanics

The results obtained from the basalt samples (locality 2A) are illustrated in Figure 6. The intensity of NRM was low (for basalts) at the start (10^{-4} emu/cc), and generally decreased to less than 10 percent of the original after 200 oe AF demagnetization. Although within-sample grouping was fair, the between-sample scatter was considerable, and no consistent average could be obtained despite demagnetization to 700 oe. The data obtained from these samples are omitted from further consideration.

Redbeds

The final results obtained after thermal demagnetization to 550°C (Fig. 7) of three specimens from each sample are illustrated in Figure 8 and listed in Tables 2 and 3. Two of the 29 samples gave highly scattered results ($k < 3$) and have been omitted. Single samples at both localities 1 and 2 have remanence orientations almost diametrically opposed to the majority (Table 2). The orientation of these anomalous samples is considered reliable, and their presence signifies the occurrence of paleomagnetic reversals at both localities. However, the two samples have been omitted from the respective vector averages shown in Table 3.

The vectors erased by thermal demagnetization between NRM and 200°C, and between 200 and 550°C are shown in Figure 9. The former are quite distinct and tend toward the present earth's field. The latter are very similar to the stable remanence remaining in terms of declination (southwest), but are inclined both above and below the horizontal.

Discussion

Locality 1 (Mérida facies).—As is evident in Table 3 and Figures 8 and 10, a considerable difference exists between samples from the two localities with respect to both orientation and

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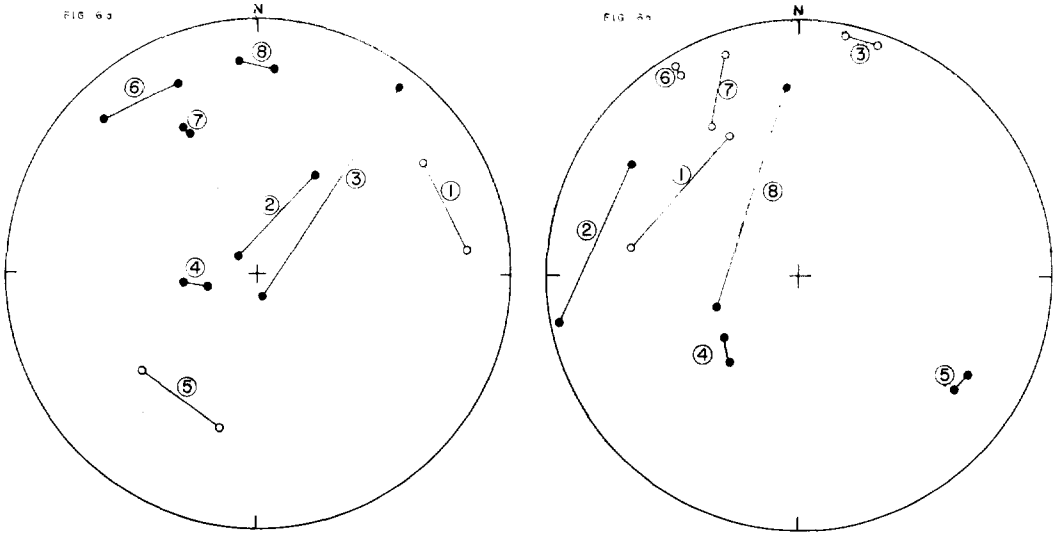


FIG. 6.—Equal area projection showing paleomagnetic data obtained from basalt samples: **(a)** NRM; **(b)** after 300-oe AF demagnetization. Bar joins vectors of specimens from same sample; dots, north seeking pole on lower hemisphere; circles, on upper hemisphere. Circled figures are specimen numbers.

intensity of remanent magnetization. The paleomagnetism of locality 1 (Mérida) is *not* the reverse of locality 2 (Sabaneta), and its similarity to the present field (Table 2) suggests that it was acquired in Quaternary or Holocene time, approximately parallel with the ambient field. An alternative explanation may be invoked involving rotation of paleomagnetic vectors from an original northeast declination as a result of structural rotation around a vertical axis (see discussion on structural geology). This is considered highly unlikely. It is believed that no greater paleomagnetic significance can be assigned to these results.

Locality 2 (Sabaneta facies).—The results from locality 2 are moderately consistent and quite distinct from the present earth's field. Of

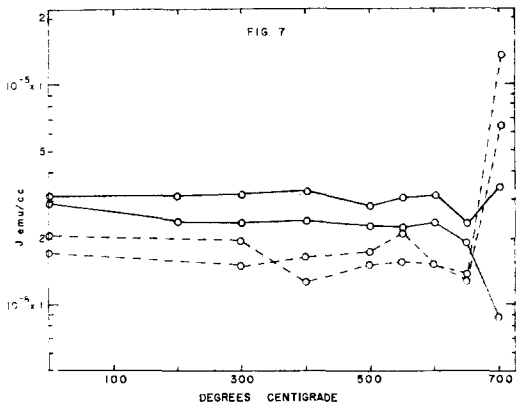


FIG. 7.—Representative intensity (J) versus temperature curves obtained on thermal demagnetization of redbed samples. Broken lines, locality 1 (Mérida); solid lines, locality 2B (Sabaneta).

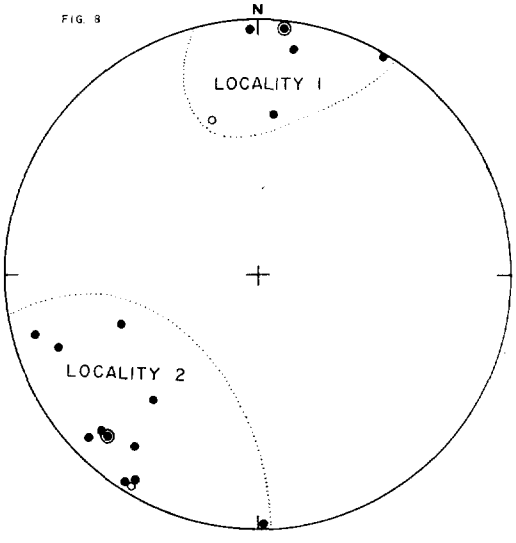


FIG. 8.—Equal area projection showing mean site vectors from localities 1 and 2B after thermal demagnetization to 550°C. Uncorrected for bedding attitude. Legend as for Figure 6. Encircled dots are mean vectors for each locality.

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Table 2. Paleomagnetic Data for Redbed Samples*

	No.	N	n	NRM				550°C			
				D	I	k	α95	D	I	k	α95
	201	1	3	16	9	(28	24)	9	12	(16	32)
	202	1	3	340	65	(5	66)	5	37	(7	53)
Locality 1	203	1	3	47	7	(26	25)	30	00	(40	20)
	204	2	5	339	−25	20	58	343	−37	38	42
	205	1	3	357	8	(1251	4)	358	5	(2,700	2)
Reversed Sample	203	1	3	183	−4	(13	36)	193	2	(19	29)
	1	2	6	251	44	67	31	211	−4	5	—
	2	1	3	214	35	(46	18)	226	9	(54	17)
	5	1	3	229	23	(3	82)	220	36	(12	37)
	6	1	3	278	62	(25	25)	250	42	(6	58)
	8	1	3	232	27	(18	30)	211	7	(47	18)
Locality 2	9	1	3	252	54	(219	8)	213	4	(93	13)
	10	1	3	187	32	(30	23)	179	3	(88	13)
	11	4	12	265	34	28	18	250	17	59	12
	12	2	6	289	47	4	90	255	10	22	56
	13	1	3	229	33	(42	19)	225	14	(6	56)
	14	4	12	230	38	21	20	216	18	88	10
Reversed Sample	6	1	3	37	−21	(28	24)	28	−12	(15	34)

* Site averages for NRM and after thermal demagnetization to 550°C. Uncorrected for bedding attitude. *No.*, site number; *N*, number of samples; *n*, number of cores measured; *D*, declination; *I*, inclination; *k*, Fisher's dispersion parameter; *α95*, half angle of cone of 95 percent confidence, Where there is only one sample at a site, statistical parameters given (enclosed in parentheses) are for the sample, and not strictly comparable with other site-means.

the 20 samples from 11 sites, one (from site 6) was reversed from the norm. The field and laboratory reliability of the orientation of this "reversed" sample was checked as far as possible, and additional cores gave identical results.

Although samples from both limbs of an anticline were collected so that a fold test could be made, only one (Site 2) from the northwest-dipping limb survived transport and drilling (the others disintegrated). The rest of the samples are from the southeast limb. It so happens, however, that the remanence declination is very close to the strike of the beds, so that correction for bedding attitude has very little effect on the orientations. Nevertheless, there is such diversity of bed attitudes that Fisher's k decreases from 11 to 7 when the bed correction is made (*i.e.*, the scatter increases). Although not quite significant at the 95 percent confidence level (McElhinny, 1964), the change is such as to suggest that the magnetism is secondary, and acquired after the folding.

Age of Magnetism

The Sabaneta redbeds were folded tightly in pre-La Quinta time, probably in Middle to Late Permian. Subsequently, together with the La Quinta Formation, the older redbeds were subject to early Tertiary and later pulses of uplift.

It is possible that the acquisition of the secondary remanence was associated with the Late Permian diastrophism. It is entirely feasible that the metamorphism, however slight, associated with the intense folding, could result in the acquisition of a chemical remanence. However, the pole position calculated from the magnetic

Table 3. Mean Vectors for Two Localities, After 550°C Thermal Demagnetization*

		In Situ					Corrected for Bed Attitude			
		n	D	I	k	α95	D	I	k	α95
Locality 1	NRM	5	7	13	4	44	358	10	4	47
	550°C	5	6	4	7	31	358	10	6	33
Locality 2	NRM	11	238	42	11	14	215	42	8	17
	550°C	11	223	15	11	15	216	10	7	18
La Quinta Type III Locality	550°C Gr. III-1	15	36	−11	11	12	44	2	8	14
	550°C Gr. III-2	3	210	5	39	20	213	0	28	24
Present dipole field			00	16						
Geomagnetic field			01	37						

* With and without correction for bedding attitude; *n*, number of sites in average; other symbols as in Table 2. For comparison are shown mean vector obtained for the La Quinta redbeds at type locality (Hargraves and Shagam, 1969) and present dipole and geomagnetic field at the sampling site.

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data is 45S, 144W, which is considerably different from the Permian pole position for South America reported by Creer (1965) at 65S, 13W. The paleolatitude of the sampling site is approximately the same for both poles, but the declination is different by 60°. Although we have already postulated a clockwise tectonic rotation of the Andean block, based on the La Quinta paleomagnetic data (Hargraves and Shagam, 1969), a rotation of this magnitude seems unreasonable.

If the remanence is Permian in age, then for an *in situ* Permian orientation, correction should be made for the Tertiary tilting. As outlined previously (Fig. 5), no such additional rotation of the Paleozoic redbeds is considered to have occurred in the post-Paleozoic tilting. Furthermore, if such correction is made, the discrepancy between the Sabaneta pole and Creer's Permian pole increases rather than decreases. The evidence thus does not favor a Permian age for the Sabaneta paleomagnetism.

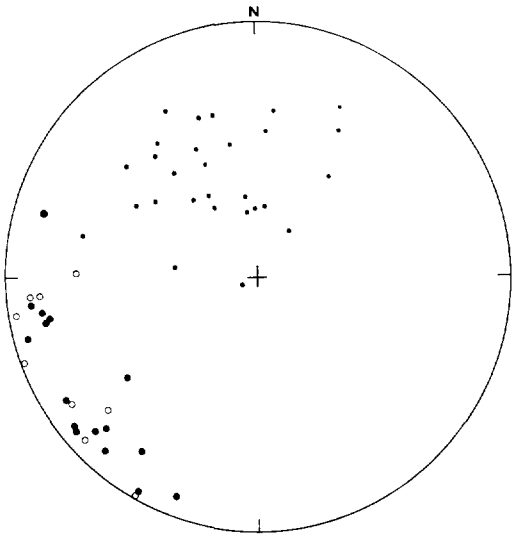


FIG. 9.—Vectors erased on thermal demagnetization of specimens from locality 2. Small dots = NRM — 200°C; large dots and circles = 200–550°C. Legend as for Figure 6.

The Sabaneta paleomagnetic results resemble those obtained from the La Quinta sandstones, and in particular are almost identical with those obtained from the La Quinta type section (locality III, Hargraves and Shagam, 1969; Table 1). The La Quinta type section samples had mostly northeast declination, and only a few southwest, whereas the reverse is true in the Sabaneta. However, it should be noted that

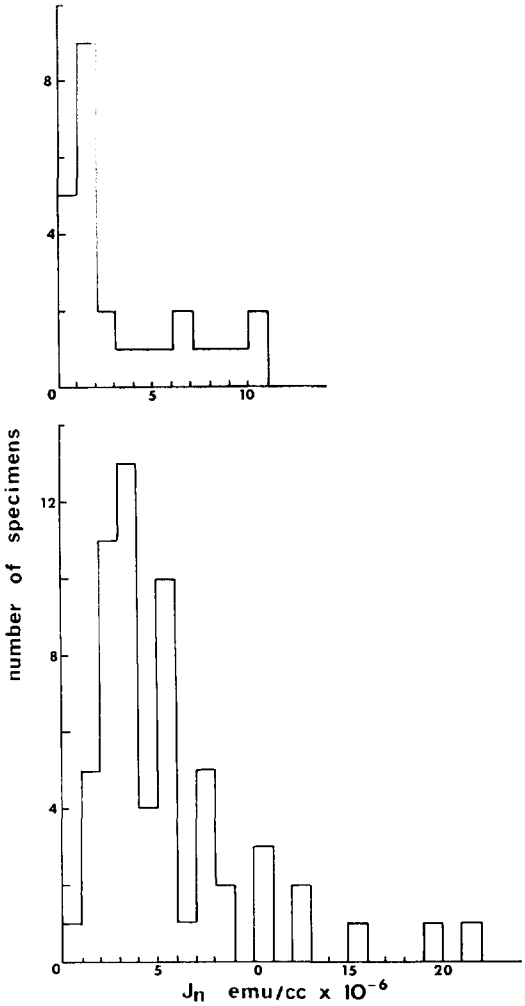


FIG. 10.—Histogram comparing variation in NRM intensity of redbed specimens at two localities. Upper figure, locality 1, n = 25; lower figure, locality 2B, n = 70.

Creer's (1962) sampling in the La Quinta type area (not precisely the same locality as our III) produced a preponderance of southwest declinations and positive inclinations. In any case, the magnetic axis in both is almost identical (Table 3).

Conclusions

These data favor the conclusion that the present remanent magnetism of the Sabaneta redbeds at the type locality was acquired almost contemporaneously with that of the La Quinta Formation and was associated with the uplift of the Andes in Tertiary time.

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The southwest declination and positive inclination would place the sampling site in the southern paleomagnetic hemisphere at the time. This is anomalous in that the available data for South America (Creer, 1965) would place Venezuela in the northern hemisphere at least since the Early Cretaceous on the basis of the paleomagnetic characteristics of the Sierra Geral lavas (120–135 m.y.). As was concluded for the La Quinta, however, the magnetism was acquired during the Tertiary uplift, when north- or southward tilting may have occurred. The oscillation above and below the horizontal of the vector erased between 200 and 500°C (Fig. 9) is in keeping with this interpretation. At present this seems the simplest explanation for the anomaly.

Consistent with the La Quinta data, the Sabaneta declination is distinctly NNE-SSW. Also, if compared with the available data for South America (Creer, 1965), Venezuela (and the continent as a whole) has not had a declination in the northeast-southwest quadrants since Pennsylvanian time.

The Sabaneta data confirm the declination anomaly, and as was suggested to account for the La Quinta results, a tectonic rotation of the Andean block subsequent to the acquisition of magnetization is the most plausible explanation.

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